

FLEETIQ GUARDIAN

Remote Driver Intelligence & Collision Risk Platform

Challenge #3: Driver Intelligence Platform
Challenge #1 scoring + Challenge #2 TTC as core engines

10

BLIND-TEST TRIPS
RANKED

17,999

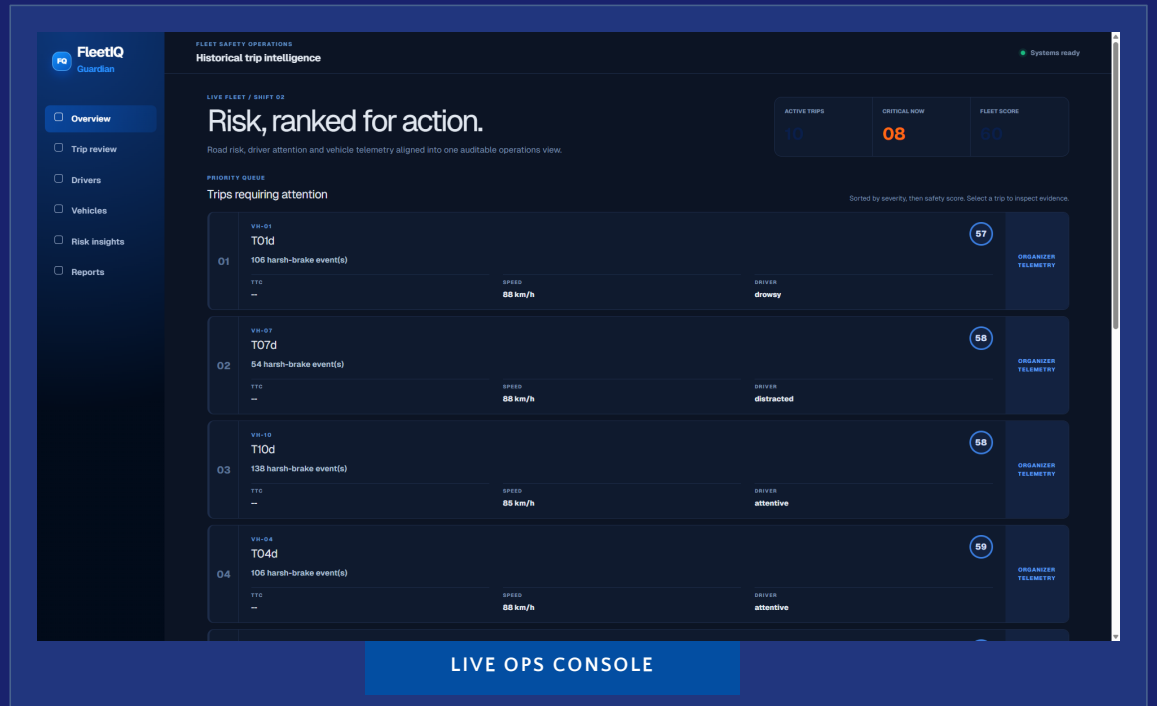
CUSTOM YOLO LABELS

1

AUDITABLE
RISK VIEW

Team UchiHahaha | Phi, Trung, Dung, Kha, Tu

Live local dashboard screenshot, build v1.0.0



KHONG CHI LA AI: DAY LA WORKFLOW CO BANG CHUNG

Mỗi claim được map tới output, route/API, artifact và limitation để BGK kiểm tra nhanh.

01 INNOVATION

Fusion road risk + driver state + telemetry into one evidence-first decision.

02 FEASIBILITY

Custom detector, deterministic fusion, Docker Compose and static artifact fallback.

03 TECHNICAL DEPTH

Stereo/depth TTC proxy, DMS state, event schema, API replay and CSV output.

04 BUSINESS VALUE

Fleet ranking, incident review and targeted coaching instead of raw video review.

REVIEW PATH: dashboard → risky trip → synchronized evidence → score explanation → coaching context

5 LUONG TIN HIEU, 1 QUYET DINH VAN HANH

Challenge #3 dùng trọn bộ dữ liệu thay vì biến một model đơn lẻ thành sản phẩm.

ROAD

Stereo road cameras

Object + context

DRIVER

In-cabin camera

Attention state

DEPTH

Depth + calibration

Distance / TTC

TELEMETRY

Vehicle signals

Speed + handling

LABELS

GT / fusion metadata

Validate + align

17,999

CUSTOM ROAD-OBJECT LABELS

10

SCORED TRIPS IN FLEET VIEW

1,800

FRAMES PER SUBMISSION FILE

5

ACCEPTED DRIVER STATES

Inputs: organizer starter kit; output contract: predictions/UchiHahaha/T01d.csv ... T10d.csv

RAW VIDEO KHONG PHAT SINH QUYET DINH AN TOAN

Fleet manager cần biết xe nào rủi ro, vì sao, lúc nào và hành động tiếp theo là gì.

01

DATA PHAN MANH

Camera, DMS, depth và telemetry nằm ở luồng riêng; reviewer không thể kể lại sự kiện.



02

ALERT THIEU NGU CANH

Một TTC đơn lẻ hoặc một frame DMS không đủ để đánh giá mức độ rủi ro.



03

KHONG CO EVIDENCE

Không có timestamp, score impact và visual evidence thì coaching không thể audit.



FLEETIQ BIEN 'NHIEU TIN HIEU' THANH 'MOT QUYET DINH CO THE KIEM TRA'

RISK, RANKED FOR ACTION

Trang đầu tiên là operations console: đội xe được xếp theo severity và safety score, không phải landing page.

LIVE FLEET VIEW

Một màn hình để chọn đúng trip cần xem trước.

Mỗi card nổi score, speed, driver state và trigger sự kiện. Reviewer có thể drill-down trực tiếp vào evidence.

10

ACTIVE TRIPS

08

CRITICAL NOW

60

FLEET SCORE

FleetIQGuardian

Overview

Trip review

Drivers

Vehicles

Risk insights

Reports

FLEET SAFETY OPERATIONS

Historical trip intelligence

LIVE FLEET / SHIFT 02

Risk, ranked for action.

Road risk, driver attention and vehicle telemetry aligned into one auditable operations view.

PRIORITY QUEUE

Trips requiring attention

Sorted by severity, then safety score. Select a trip to inspect evidence.

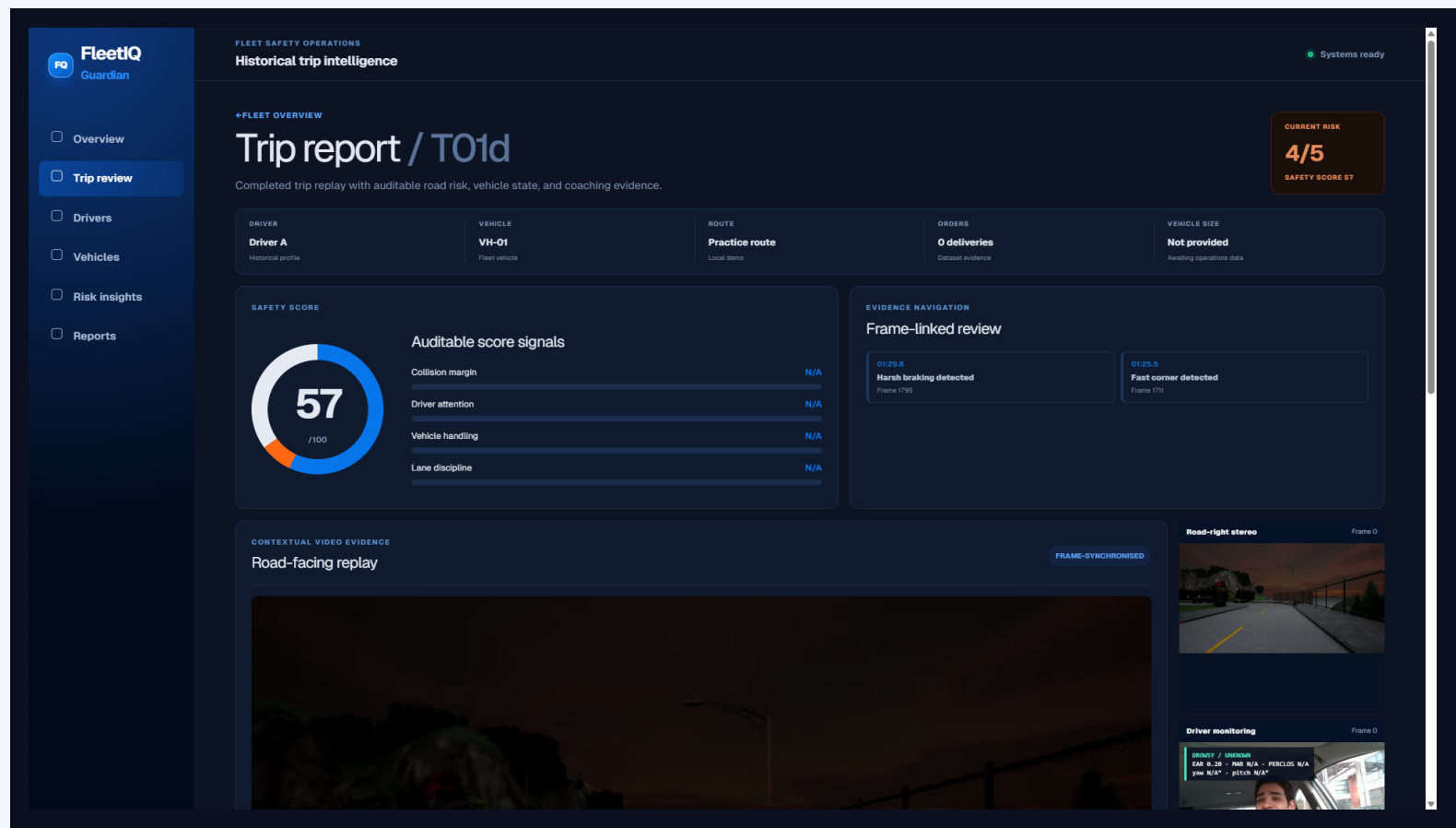
01	VH-01 T01d 106 harsh-brake event(s) TTC SPEED 88 km/h DRIVER drowsy 57 ORGANIZER TELEMETRY
02	VH-07 T07d 54 harsh-brake event(s) TTC SPEED 88 km/h DRIVER distracted 58 ORGANIZER TELEMETRY
03	VH-10 T10d 138 harsh-brake event(s) TTC SPEED 85 km/h DRIVER attentive 58 ORGANIZER TELEMETRY
04	VH-04 T04d 106 harsh-brake event(s) TTC SPEED 88 km/h DRIVER attentive 59 ORGANIZER TELEMETRY

Systems ready

Live local Docker dashboard screenshot: <http://localhost:3000/> | build v1.0.0

T01d: SCORE -> EVENT -> EVIDENCE

Trip report giữ evidence navigation, replay và score explanation trên cùng luồng review.



Live local Docker dashboard screenshot: /trips/T01d | actual UI state is shown, including any N/A detail fields.

57/100 SAFETY SCORE

Risk card gives a reviewer one starting signal.

4/5 CURRENT RISK

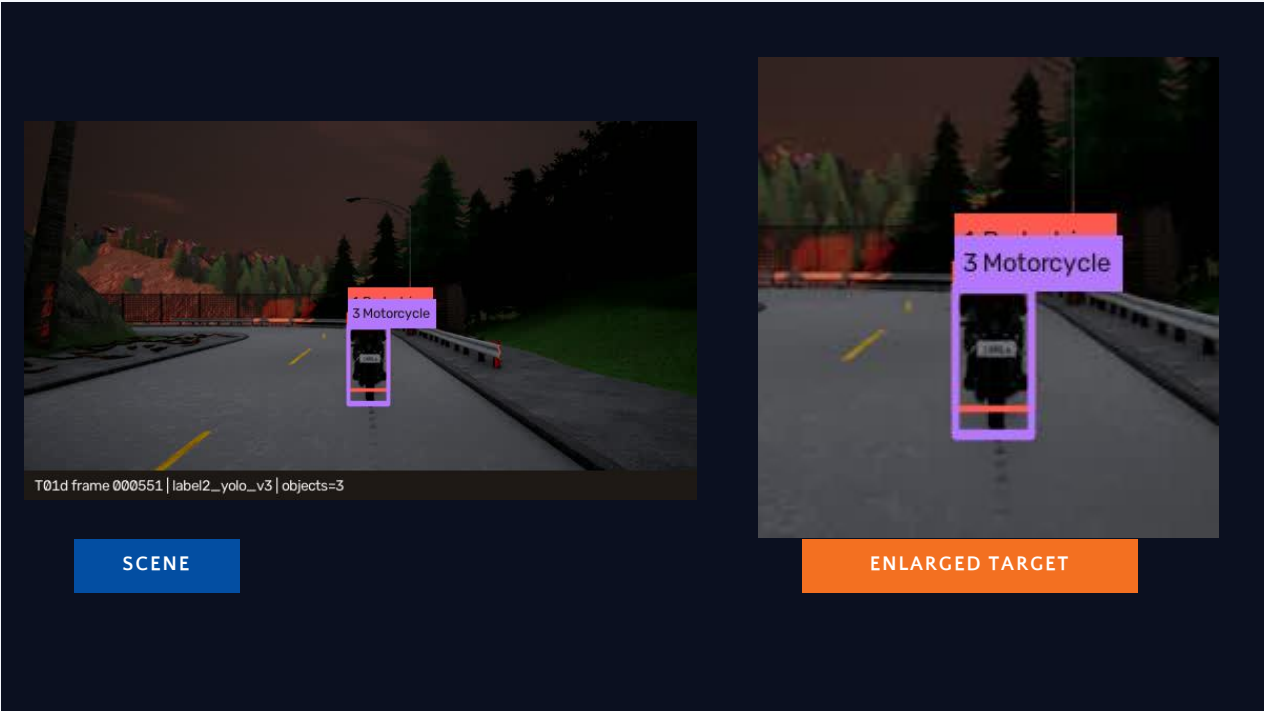
Severity is traceable to road, driver and telemetry artifacts.

2 FRAME-LINKED EVENTS

Harsh brake and fast corner point to specific frame ranges.

MOTORCYCLE EVIDENCE, NOT A BLACK-BOX ALERT

Frame 551 turns a model output into an inspectable road-risk record.



Motorcycle

confidence 0.3941

5.02 m

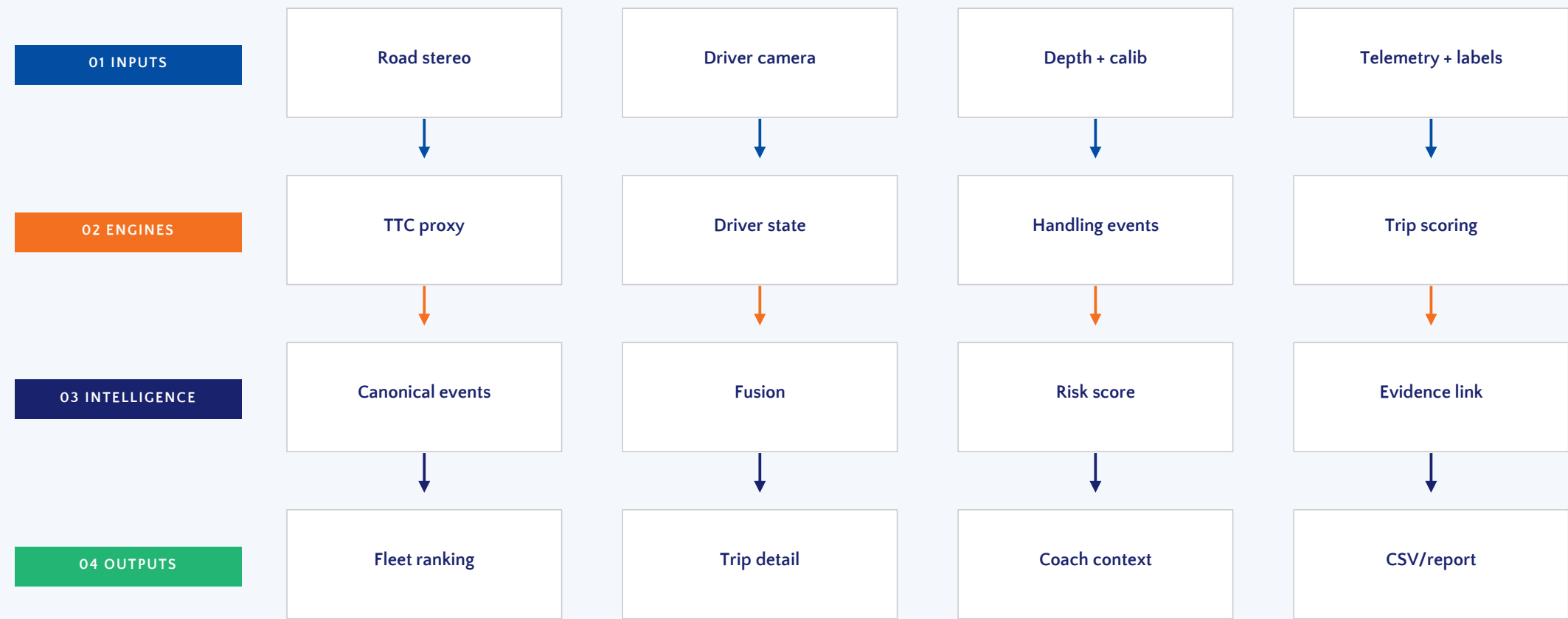
depth ROI estimate

DISCLOSURE

The same frame has an overlapping pedestrian false positive. We keep it visible rather than hide it.

FROM SENSORS TO A FLEET ACTION

The intelligence layer carries one event schema: severity, confidence, score impact, explanation and evidence.



Runtime: precomputed artifacts -> FastAPI -> Next.js | Workers use bounded JSON/JSONL and compose profiles.

IMPROVE ERROR AND RECALL; DISCLOSE THE TRADE-OFF

Organizer evaluation on full-GT T01-Sample only. This is not a blind-test score for T01d-T10d.

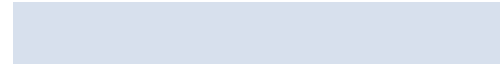
CRITICAL TTC MAE

58.595s

Organizer SGBM baseline

19.612s

Custom detector + depth ROI



66.5% lower critical-zone MAE

CRITICAL TTC F1

0.125

BASELINE

0.240

CUSTOM

TTC composite: 30.6 baseline vs 28.3 custom

Reason: custom path has higher false-positive rate (0.058 vs 0.008).

BTC CUNG CAP DATA; UCHIHAHAHA CUNG CAP WORKFLOW

Giá trị không nằm ở việc gọi một model, mà ở integration để Fleet Manager hành động được.

STARTER / BASELINE

- Road, driver, depth, calibration, telemetry
- Team-kit fixed-ROI StereoSGBM baseline
- Redacted scored-trip ground truth
- YOLOP road/lane segmentation

TEAM-OWNED

- 17,999 custom YOLO v3 labels + selected checkpoint
- Road/DMS/fusion artifacts and depth-TTC evidence
- Fleet ranking, replay, video-range and overlay routes
- Submission exporter, validator, evaluation evidence and package

REMOVE OUR INTEGRATION -> raw data remains, but the reviewer loses ranked risk, synchronized evidence and auditable CSV outputs.

DEMO HAY HANDOFF: CUNG MOT BUILD, CUNG MOT EVIDENCE

Release v1.0.0 separates source, organizer data and runtime artifacts so reviewers can rerun the final workflow.

RUN

Docker Compose

API, web, MinIO, Redis, PostgreSQL, MQTT, model mock and CarSky bridge.

REVIEW

Evidence package

Models, trip artifacts, evaluator reports, predictions and SHA-256 manifest.

SCALE

OEM / fleet path

Input, engine, intelligence and output layers can expand without replacing the workflow.

TEAM OWNERSHIP

Phi: roadface + automotive | Trung: DMS | Dung: agent/coaching | Kha: CV/ML + depth | Tu: backend + dashboard

CarSky disclosure: bridge is part of the build; Android Automotive end-to-end HMI must be demonstrated separately before claiming it.

REVIEW IN 3 MINUTES

FROM RISK QUEUE TO COACHING CONTEXT

- 01 Open fleet ranking
- 02 Select T01d
- 03 Inspect timeline + replay
- 04 Open frame-linked evidence
- 05 Assign targeted coaching

FINAL DISCLOSURE

- Redacted trips have no local ground truth.
- Practice evaluation is not blind-test accuracy.
- Frame 551 false positive is kept visible.
- CarSky HMI remains partial until recorded.

v1.0.0

Source + runtime handoff
for reproducible review

THANK YOU

Release runbook: [docs/runbooks/final-release.md](#)